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Portfolio: knocktern.github.io/Portfolio

Summary

Final-trimester Computer Science and Engineering student at United International University with hands-on full stack web development experience building database-driven applications using Python, Flask, MySQL, and REST APIs. Practical background in IoT systems with ESP32 microcontrollers and real-time communication protocols. Strong analytical and troubleshooting skills, with a keen interest in EV charging technology, smart mobility, and applied software integration. Comfortable working in both office and field environments.

Education

United International University (UIU)

Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering, CGPA: 3.83

Expected Graduation: October 2026

Currently in final trimester

Technical Skills

Programming Languages: Python, JavaScript, C, C++, Java

Web Development: Flask, REST APIs, JSON, HTML/CSS, WebRTC, Socket.IO

Databases: MySQL, SQL, SQLAlchemy

IoT & Systems: ESP32, sensor integration, Unix-like OS kernel development

Machine Learning: Scikit-learn, Pandas, NumPy, OpenCV, Matplotlib, Seaborn

Tools & Version Control: Git, GitHub, VS Code, Jupyter

Projects

HireMe – Job Matching System

Flask, MySQL, Python

1st Runner Up, Software Engineering Lab Category, UIU CSE Project Show (Fall '25)

- Developed a full stack web platform connecting job seekers with employers through skill-based matching algorithms
- Built role-based access for Admins, Employers, and Job Seekers with distinct permission tiers
- Implemented advanced multi-parameter search and filtering by location, experience, skills, and salary using SQL queries
- Designed the platform around real hiring and interview workflows, recognized at the UIU CSE Project Show
- GitHub: github.com/Knocktern/HireMe

The Quiet Game

JavaScript, WebRTC, Socket.IO

Judges Choice Award Winner, UIU HackDay 2026 Hackathon

- Built a real-time multiplayer web application using sign language recognition
- Integrated WebRTC peer-to-peer video streaming and Socket.IO for live multiplayer state synchronization
- Developed live leaderboard functionality with real-time score updates across connected clients
- GitHub: github.com/Knocktern/The-Quiet-Game

Smart Car Parking System

ESP32, IoT, C++

Electronics Course Project, Summer 2024

- Developed an IoT-based automated parking management system on ESP32 microcontroller hardware
- Integrated ultrasonic sensors for vehicle detection and servo motor control for automated barrier operation
- Implemented real-time LED status indicators displaying live parking slot availability

Hospital Management System

Flask, MySQL, SQLAlchemy, REST

- Engineered a full stack Flask web application managing doctor and patient records, appointment scheduling, and medical history
- Designed and integrated a relational MySQL database via SQLAlchemy ORM, coordinating API and database operations
- Implemented role-based access control for Managers, Doctors, and Patients with automated scheduling conflict detection
- Automated email notification workflows using Flask-Mail and APScheduler for appointment reminders
- GitHub: github.com/Knocktern/Hospital_Management_System_DBMS-

Pothole Detection and Severity Assessment

Python, YOLOv8, OpenCV

Digital Image Processing Course Project

- Built a pothole detection and segmentation pipeline using a YOLOv8n-seg model for automated road damage identification
- Designed a custom Risk/Repair Priority Score (RPS) using OpenCV pixel intensity analysis to quantify severity
- Applied Albumentations augmentation pipelines to improve model robustness across varied road conditions

Student Dropout Prediction

Python, Scikit-learn, Pandas

- Built a classification model identifying students at risk of dropout to support early intervention
- Conducted exploratory data analysis and feature importance evaluation to surface key predictive factors
- Benchmarked multiple machine learning algorithms to determine optimal model performance
- Notebook: Google Colab

Achievements

Judges Choice Award Winner – UIU HackDay 2026 Hackathon

- Awarded for "The Quiet Game", a real-time multiplayer sign language game built with WebRTC and Socket.IO

1st Runner Up – UIU CSE Project Show (Fall '25), Software Engineering Lab Category

- Recognized for "HireMe", a full stack platform built around real hiring and interview workflows

Joint 2nd Runner Up – UIU CSE Project Show (Fall '25), FYDP-1 Category

- Awarded for a research-based final year design project

Co-Curricular Activities

Volunteer, Study World Roadshow

April 2026

IDP Education Bangladesh

Hotel Sheraton, Dhaka

- Supported event coordination and guided participants at IDP's Study World Roadshow
- Received a Certificate of Appreciation from IDP Education Bangladesh for contribution to the event